

Performance Results for Clamped PID Controller (Torque Limit ± 2.0)

Performance at Noise Level $\sigma = 0.0$

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	0.190 ± 0.000	0.000 ± 0.000	0.0030 ± 0.0000
-0.9443	0.276 ± 0.000	0.000 ± 0.000	0.0031 ± 0.0000
0.1559	0.063 ± 0.000	0.000 ± 0.000	0.0029 ± 0.0000
-0.6195	0.226 ± 0.000	0.000 ± 0.000	0.0030 ± 0.0000
-0.8419	0.262 ± 0.000	0.000 ± 0.000	0.0031 ± 0.0000

Performance at Noise Level $\sigma = 0.001$

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	0.190 ± 0.000	0.001 ± 0.000	0.0031 ± 0.0000
-0.9443	0.276 ± 0.000	0.001 ± 0.000	0.0032 ± 0.0000
0.1559	0.064 ± 0.001	0.001 ± 0.000	0.0030 ± 0.0000
-0.6195	0.226 ± 0.000	0.001 ± 0.000	0.0031 ± 0.0000
-0.8419	0.262 ± 0.000	0.001 ± 0.000	0.0032 ± 0.0000

Performance at Noise Level $\sigma = 0.01$

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	0.531 ± 0.224	0.120 ± 0.009	0.0372 ± 0.0006
-0.9443	0.472 ± 0.426	0.121 ± 0.006	0.0391 ± 0.0063
0.1559	0.366 ± 0.384	0.129 ± 0.007	0.0388 ± 0.0035
-0.6195	0.731 ± 0.416	0.129 ± 0.037	0.0399 ± 0.0011
-0.8419	0.210 ± 0.009	0.115 ± 0.003	0.0388 ± 0.0042

Performance at Noise Level $\sigma = 0.1$

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	N/A	1.119 ± 0.352	N/A
-0.9443	N/A	0.874 ± 0.192	N/A
0.1559	N/A	0.954 ± 0.256	N/A
-0.6195	N/A	0.973 ± 0.217	N/A
-0.8419	N/A	0.867 ± 0.089	N/A

Analysis and Discussion

The following analysis details the performance of the PID controller ($K_p=2$, $K_i=0$, $K_d=0.3$) for the inverted pendulum, with its output torque clamped to a maximum of ± 2.0 . The results show a controller that is highly effective and stable in low-noise conditions but demonstrates a clear and rapid degradation in performance as noise increases, culminating in system failure.

Baseline Performance (Low Noise: $\sigma = 0.0$ and 0.001)

In the noise-free and low-noise environments, the controller's performance is excellent and highly predictable.

- **No Overshoot:** The most notable characteristic is the **zero overshoot** in the noise-free runs. The metric for overshoot only registers a value after the pendulum crosses the zero line. This result indicates the controller is tuned for a **critically damped or overdamped response**, smoothly bringing the pendulum to the vertical position without oscillating past the target. This prioritizes stability over raw speed.
- **Fast and Stable:** The stabilisation times are quick and directly proportional to the initial angle, which is expected behavior.
- **Excellent Noise Immunity:** The introduction of minimal noise ($\sigma = 0.001$) has a negligible impact. The performance remains almost identical to the ideal case, demonstrating robustness to minor sensor imperfections.

The Performance "Elbow" (Medium Noise: $\sigma = 0.01$)

This noise level represents the critical turning point where the controller begins to struggle significantly. While the system remains stable, its performance becomes erratic and unreliable.

- **Loss of Consistency:** The most telling sign of degradation is the **explosion in standard deviation** for stabilisation time. A value like 0.472 ± 0.426 s means the time to stabilize is no longer predictable; in some runs the controller gets lucky, while in others it struggles for much longer.
- **Emergence of Overshoot:** The controller can no longer maintain its smooth, no-overshoot trajectory. It is now actively "fighting" the noise, resulting in an average overshoot of about 0.12 radians (~7 degrees).
- **Degraded Precision:** The steady-state error increases by a full order of magnitude to approximately 0.039 rad. The controller cannot hold the pendulum perfectly still, instead allowing it to oscillate within a much wider band.

System Failure (High Noise: $\sigma = 0.1$)

At this high noise level, the controller is overwhelmed and can no longer perform its function.

- **Complete Failure to Stabilize:** The system **fails to meet the stability criteria** in every single run across all initial angles.

- **Destabilizing Control Action:** The overshoot is now enormous and unpredictable, often exceeding the initial angle. This indicates that the controller's actions, corrupted by the high-frequency noise (especially in the Derivative term), are making the system *less* stable. The limited torque of 2.0 is insufficient to counteract these powerful, noise-induced disturbances.

Conclusion and Tuning Insights

This PID controller is tuned for **high precision and a stable, non-oscillatory response in ideal conditions**, but it sacrifices robustness to do so.

1. **Lack of Integral Action ($K_i=0$):** The small but persistent steady-state error even at $\sigma=0$ suggests the controller cannot compensate for minor constant disturbances. A small K_i term could potentially eliminate this.
2. **Sensitivity to Noise:** The failure at $\sigma=0.1$ shows a classic weakness. The Derivative (K_d) term, while crucial for dampening, is also highly sensitive to noise, which leads to erratic control output.
3. **Limited Authority:** The torque limit of 2.0 is the ultimate bottleneck. While sufficient for ideal conditions, it prevents the controller from making the strong corrections needed to overpower the noise at higher levels.

To improve this controller, the next steps would be to **introduce signal filtering** to clean the sensor data before it reaches the PID logic or to **re-tune the gains** (e.g., reduce K_d and introduce a small K_i) to find a better balance between responsiveness and noise immunity.